The International Journal of Transfusion Medicine
Vox Sanguinis

Vox Sanguinis (2010) **99**, 212–219

Vox Sanguinis © 2010 International Society of Blood Transfusion
No claim to original US government works
DOI: 10.1111/j.1423-0410.2010.01340.x

ORIGINAL PAPER

Reduced risk of transfusion-transmitted HIV in Kenya through centrally co-ordinated blood centres, stringent donor selection and effective p24 antigen–HIV antibody screening

S. V. Basavaraju,[1,2] J. Mwangi,[3] J. Nyamongo,[4] C. Zeh,[5] D. Kimani,[3] R. W. Shiraishi,[6] R. Madoda,[4] J. A. Okonji,[7] W. Sugut,[4] S. Ongwae,[4] J. P. Pitman[1] & L. H. Marum[1]

[1] *HIV Prevention Branch, Global AIDS Program, National Center for HIV/AIDS, viral Hepatitis, STD and TB Prevention, U.S. Centers for Disease Control and Prevention, Atlanta, GA, USA*
[2] *Epidemic Intelligence Service, U.S. Centers for Disease Control and Prevention, Atlanta, GA, USA*
[3] *Global AIDS Program, U.S. Centers for Disease Control and Prevention, Nairobi, Kenya*
[4] *Kenya National Blood Transfusion Service, Nairobi, Kenya*
[5] *Centers for Disease Control and Prevention (CDC-Kenya), Kisumu, Kenya*
[6] *Epidemiology and Strategic Information Branch, Global AIDS Program, National Center for HIV/AIDS, viral Hepatitis, STD and TB Prevention, U.S. Centers for Disease Control and Prevention, Atlanta, GA, USA*
[7] *Centre for Global Health Research, Kenya Medical Research Institute, Kisumu, Kenya*

Background Following a 1994 study showing a high rate of transfusion-associated HIV, Kenya implemented WHO blood safety recommendations including: organizing the Kenya National Blood Transfusion Service (NBTS), stringent blood donor selection, and universal screening with fourth-generation p24 antigen and HIV antibody assays. Here, we estimate the risk of transfusion-associated HIV transmission in Kenya resulting from NBTS laboratory error and consider the potential safety benefit of instituting pooled nucleic acid testing (NAT) to reduce window period transmission.

Methods From November to December 2008 in one NBTS regional centre, and from March to June 2009 in all six NBTS regional centres, every third unit of blood screened negative for HIV by the national algorithm was selected. Dried blood spots were prepared and sent to a reference laboratory for further testing, including NAT. Test results from the reference laboratory and NBTS were compared. Risk of transfusion-associated HIV transmission owing to laboratory error and the estimated yield of implementing NAT were calculated.

Findings No cases of laboratory error were detected in 12 435 units tested. We estimate that during the study period, the percentage of units reactive for HIV by NAT but non-reactive by the national algorithm was 0·0% (95% exact binomial confidence interval, 0·00–0·024%).

Interpretation By adopting WHO blood safety strategies for resource-limited settings, Kenya has substantially reduced the risk of transfusion-associated HIV infection. As the national testing and donor selection algorithm is effective, implementing NAT is unlikely to add a significant safety benefit. These findings should encourage other countries in the region to fully adopt the WHO strategies.

Key words: blood transfusion, HIV, Kenya, nucleic acid testing.

Received: 2 March 2010,
revised 9 April 2010,
accepted 10 April 2010,
published online 28 June 2010

Correspondence: Dr Sridhar V. Basavaraju, U.S. Centers for Disease Control and Prevention, Atlanta, GA 30341, USA
E-mail: etu7@cdc.gov
This work has been submitted as an abstract to the XVIII International AIDS Conference, Vienna, Austria (July 18–23, 2010).

This article was published online on 19 May 2010. An error was subsequently identified. This notice is included in the online and print versions to indicate that both have been corrected 13 July 2010

Introduction

Unsafe blood transfusions have historically contributed a substantial burden of human immunodeficiency virus (HIV) infections in sub-Saharan Africa [1]. Kenya in particular has received attention for the role of unsafe blood donations and inadequate laboratory screening in transmission of HIV via blood transfusion. In 1994, Moore *et al.* found a 6·4% prevalence of HIV among Kenyan blood donors and estimated that 2·0% of transfusions in the study population transmitted HIV [2]. The reasons for unsafe blood transfusions in Kenya were similar to those found elsewhere in sub-Saharan Africa. Blood safety was plagued by numerous structural weaknesses including deficiencies in governmental regulations and financing and limited external donor support. Most blood units were collected from family, replacement or paid donors, individuals often at high behavioural risk for HIV infection [2]. Behavioural screening of potential donors was not routinely conducted, and a formal deferral system did not exist to exclude individuals at high risk for HIV [2]. Additionally, laboratory testing for HIV was not quality assured. Testing, as well as recording and reporting of results, did not follow standardized methodologies. Laboratory technicians were inadequately trained and often held incorrect perceptions about risks of transfusion-associated HIV transmission. Improper or incomplete screening of donated blood was exacerbated by stock-outs of laboratory kits and reagents. When performed, screening was conducted with antibody-only-based enzyme-linked immunoadsorbant assays (ELISA) that were unreliable in detecting acute, pre-seroconversion, 'window period' HIV infections. Lastly, inappropriate or unnecessary transfusions contributed to chronic blood shortages, thus perpetuating a cycle of unsafe collections to meet clinical demand [2].

In 2002, WHO released an Aide-mémoire for National Blood Programmes [1]. This comprehensive strategy encouraged resource-limited countries to invest in four key elements: (1) The establishment of a nationally co-ordinated blood transfusion service; (2) the collection of blood from voluntary, non-remunerated donors with a low behavioural risk profile for HIV; (3) the implementation of universal blood screening for HIV and other transfusion-transmissible infections, and; (4) the reduction in unnecessary transfusions. The Kenya Ministry of Health, with support from the United States government and from the Japanese International Cooperation Agency, developed a new blood policy in 2001 and transformed the Kenya National Blood Transfusion Service (NBTS) into a nationally co-ordinated system with a national network of six blood centres and satellite distribution sites [3]. In 2004, blood safety was included in the HIV prevention strategy of the U.S. President's Emergency Plan for AIDS Relief (PEPFAR). Since then, PEPFAR has provided over $250 million in grants to strengthen national blood transfusion services in 14 resource-limited countries, including technical assistance and $20 million in direct grants to the Kenya NBTS [4]. All NBTS blood donors are now voluntary and non-remunerated. Donors must also pass a pre-donation health exam and respond to a stringent behavioural risk screening questionnaire. Training of laboratory staff has been undertaken by PEPFAR-funded technical assistance organizations, including AABB (formerly American Association of Blood Banks). Quality-assured laboratory screening is conducted on all donated units using fourth-generation p24 antigen and HIV 1 and 2 antibody tests which have shorter window periods of detection than the antibody-only-based assays. NBTS has also educated health care providers and advised hospitals on the appropriate clinical use of blood. In 2007, NBTS collected 123 787 units of blood, 99·5% from voluntary, non-remunerated blood donors, and supplied the vast majority of blood units used by hospitals and other health facilities that provide transfusion services [5].

While much progress has been made in Kenya, challenges and unresolved questions remain. Kenya is experiencing a severe, generalized HIV epidemic with a 7·1% national population prevalence estimate among adults aged 15–64 [6]. Despite the policy of collecting blood only from voluntary, non-remunerated donors, the percentage of NBTS collections reactive for HIV has ranged between 1·2% and 2·5% [5]. Most NBTS donors are students aged 16–19, an adult age group in Kenya with low HIV prevalence [6]. Blood shortages therefore often occur during school holidays. Inconsistent funding to mount blood drives outside of schools and higher HIV population prevalence in older adults pose challenges in retaining low-risk regular donors. In addition, the fourth-generation HIV screening assay used in Kenya may not detect p24 antigen or HIV antibodies during a pre-seroconversion window period of approximately 18 days after infection [7]. This raises the potential for false negative results in donors with acute or recent infections. In many industrialized countries, nucleic acid testing (NAT) has become the accepted standard for screening donor blood. Testing pooled samples with NAT may reduce the window period to approximately 11 days [8]. However, because of its high cost and technical difficulty, NAT is not currently used by the NBTS. This study aimed to evaluate the effectiveness of a centrally co-ordinated national blood service, stringent donor selection policies, and universal, quality-assured fourth-generation ELISA screening in reducing the risk of transfusion-associated HIV infections in the absence of the shorter window period of detection allowed by NAT. We quantified the residual risk of transfusion-associated HIV infection resulting from laboratory errors in screening, recording and reporting and

No claim to original US government works

evaluated the potential additional safety benefit of implementing NAT in Kenya.

Methods

Patient population and procedures

A nationally representative sample of whole blood units collected by all six NBTS regional blood transfusion centres and testing negative for HIV by the national algorithm (Eldoret, Embu, Kisumu, Mombasa, Nairobi, and Nakuru, Kenya) was analysed. Blood units are primarily collected through mobile campaigns, with a small proportion collected from walk-in donors. Blood units collected and screened by individual healthcare facilities (not by NBTS) were not included in this evaluation. We also did not include units that were required for urgent component production, including packed red-blood-cells, paediatric red-blood-cells, fresh frozen plasma, and platelets. To achieve the necessary sample size while minimizing the impact of the study quarantine period on blood supplies, every third eligible unit was systematically sampled from a random ordering of blood units collected each day. The study was carried out between November 2008 and June 2009. In December 2008, the study was temporarily halted as a result of nationwide blood shortages. Only one centre (Mombasa) continued sampling during this period (November–December 2008). The nationwide study resumed in March 2009 and was completed in June 2009. Sampling was also halted for a 2-week period in May 2009 in the Eldoret centre because of severe blood shortages. One traditionally low-collection month (April) was included in which HIV prevalence among collected units is usually higher because of school holidays when NBTS collects blood primarily from older age groups.

The study was approved by the Scientific Steering and Ethics Committees of the Kenya Medical Research Institute (KEMRI) and the U.S. Centers for Disease Control and Prevention's (CDC) Institutional Review Board in Atlanta, Georgia. All donors received pre-donation counselling by NBTS staff, completed the standard NBTS behavioural risk profile questionnaire, and signed a statement of understanding that all donated blood would be tested for HIV and other transfusion-transmissible infections.

Laboratory testing

Per NBTS algorithm, centres test all donated units with the fourth-generation Vironostika HIV Uni-Form II Antigen/Antibody test (bioMerieux, Inc., Durham, NC, USA). Units with non-reactive results are released into the blood supply. Reactive samples are retested with the Vironostika test and results are confirmed using the fourth-generation Murex Ag/Ab assay (Murex Diagnostics, Inc., Norcross, GA, USA). Units with reactive results with the initial Vironostika test are excluded from transfusion. Confirmatory testing is not used to clear units for transfusion, but to allow accurate return of test results. Donors with reactive results receive counselling and a referral to retesting and treatment services. These donors are also permanently excluded from future donation. All donated blood units are also screened for hepatitis B, hepatitis C, and syphilis. Only units that tested negative for HIV, hepatitis B, C, and syphilis were included in this study.

Dried blood spot (DBS) specimens were prepared from the sampled units and the units were placed in a refrigerated quarantine area pending NAT results from the reference laboratory. The quarantine period typically lasted between 4 and 7 days. DBS specimens were transported to the KEMRI/CDC Kisian Field Station reference laboratory in Kisumu, Kenya, where NAT was conducted. The Roche COBAS Ampliscreen HIV-1 test version 1.5 (Roche Diagnostic Systems, Branchburg, NJ, USA) kit was used for NAT. Prior to commencing the study, the reference laboratory adapted previously described procedures to conduct a validation trial [9–13]. This trial used the COBAS Ampliscreen assay on panels with known HIV-1 viral loads (ranging from 15 to 800 copies per 6 mm disc), and on negative controls to confirm and compare sensitivity and specificity between DBS and plasma specimens. DBS specimens did not contain personal identifiers. A unique identification number was used to transmit results back to the NBTS centres, which were responsible for linking the specimen identification numbers to patient names for donor notification purposes.

At the reference laboratory, DBS specimens were prepared into pools of 10 specimens from which nucleic acid was extracted. HIV reactive results were defined as signals equal to or greater than cut-off values. Pooled specimens with reactive results were subjected to individual NAT to identify the infected unit. To distinguish between NBTS ELISA window period and laboratory or reporting error cases, any NAT reactive DBS samples would be retested with the Vironostika kits included in the NBTS algorithm. Specimens which were non-reactive in NBTS testing, but positive by NAT and reactive by ELISA in the reference laboratory would be deemed instances of NBTS laboratory error. Specimens which tested NAT positive but were non-reactive by ELISA in the reference laboratory would be identified as units potentially detectable by NAT and falling within the ELISA window period.

Statistical analysis

Determination of sample size was based on an assumption among co-investigators that the identification of a case that is NAT reactive but ELISA non-reactive is a rarer event than

No claim to original US government works

a laboratory error. In determining the sample size, the potential number of additional infections NAT might identify was estimated. This yield was defined as the percentage of HIV-infected units falling between the 18-day ELISA window period and the 11-day NAT window period. For this estimate, a standard formula originally developed to estimate residual risk of window period HIV transmission was used [14,15]. As this formula requires an HIV incidence rate which was unavailable from NBTS records, a crude HIV incidence rate was first calculated by applying the ratio of new to existing HIV cases, using population estimates of HIV incidence and prevalence from Spectrum/AIM [16], to estimates of HIV prevalence among blood donors in Kenya. Using this method, the HIV incidence rate among Kenyan blood donors was estimated to range from 170 to 340 infections per 100 000 person-years. As this crude range included an estimate of blood donor incidence rate reported elsewhere in sub-Saharan Africa [17], the published estimate (287 infections per 100 000 person-years) was used as the primary measure. Using the estimated incidence rate and the 7-day interval between the two window periods, it was estimated that implementing NAT would potentially yield an additional 0·0055% HIV reactive units. In selecting the final sample size, Clopper–Pearson exact binomial bounds were calculated [18] for sample size ranges from 10 000 to 20 000 units, concentrating on the upper exact binomial bounds for potential additional infections detected through NAT of 0·0–0·01%. The ranges of upper exact bounds for 10 000–20 000 units were 0·030–0·015% for 0·0% and 0·056–0·036% for 0·01%.

Data analysis was performed using SAS version 9.2 (SAS Institute Inc, Cary, NC, USA). Test results from the reference laboratory were compared with results from the NBTS regional laboratories. The analysis included calculations to identify the proportion of false negative ELISA results resulting from NBTS laboratory or reporting error and the proportion and 95% exact binomial confidence intervals of additional HIV reactive blood units that would have been detected in the study period if NAT had been implemented in Kenya. Results from this study were also compared with those presented in Moore *et al.*'s report [2]. Special attention was paid to changes in the operational, regulatory, and policy environment in the 15 years between the two studies.

Results

During the study period, 40 657 blood units were collected by the six NBTS regional centres. Of these, 492 units were reactive for HIV, 1335 for hepatitis B, 441 for hepatitis C, and 79 for syphilis, and excluded from sampling. An additional 346 units were excluded owing to urgent component preparation needs. The Mombasa centre sampled 506 units in November–December 2008. Between March and June 2009, a total of 12 053 units were sampled from the six centres. A total of 12 559 DBS were sent to the reference laboratory. Of these, 124 DBS specimens were ineligible for inclusion because of defects, including mislabelling, improper preparation, heat fixation, or other contamination. A total of 12 435 DBS specimens were subjected to pooled NAT (Fig. 1).

None of the tested specimens were reactive for HIV by pooled NAT. Thus, the reference laboratory did not retest any specimens using the NBTS algorithm. The residual risk of HIV transmission by blood transfusion attributed to laboratory error in the study period was observed to be zero (Table 1). Furthermore, during the study period, the estimated percentage of all NBTS whole blood units that were reactive for HIV by NAT but non-reactive by the fourth-generation assay was 0·0% with a 95% exact binomial confidence interval of 0·00–0·024%. Table 1 summarizes improvements in national blood transfusion policies and procedures in Kenya and resulting reduction in the residual risk of HIV transmission by blood transfusion resulting from laboratory and reporting errors.

Discussion

Renewed attention to blood safety by the Ministry of Health and donor agencies has produced significant improvements since the 1994 study of Moore *et al.* [2]. Contrary to reports from across sub-Saharan Africa which have previously indicated a substantial burden of HIV infections attributed to blood transfusion [2,17,19–23], this study identified no evidence of laboratory or reporting error and only a theoretical risk of HIV transmission because of acute infections potentially detectable by NAT. These results demonstrate that a profound reduction in the risk of transfusion-associated HIV infection is possible in resource-limited countries with severe, generalized HIV epidemics, and validate the NBTS decision to implement the WHO recommendations for national blood services.

The findings of this study reaffirm the utility of fourth-generation assays in successfully screening for HIV in blood donated by individuals who have also been appropriately selected. These results further support WHO recommendations to complement HIV screening with safe donor selection practices. Kenya's focus on the selection of voluntary, non-remunerated donors coupled with behavioural and medical screening to replace the previous unregulated donor recruitment system has likely minimized the risk of window period donations. Blood services that do not conduct stringent pre-donation health and behavioural screening, and that rely on family, replacement or paid donors, are likely to have a much higher proportion of donations from individuals recently infected with HIV [24,25].

No claim to original US government works

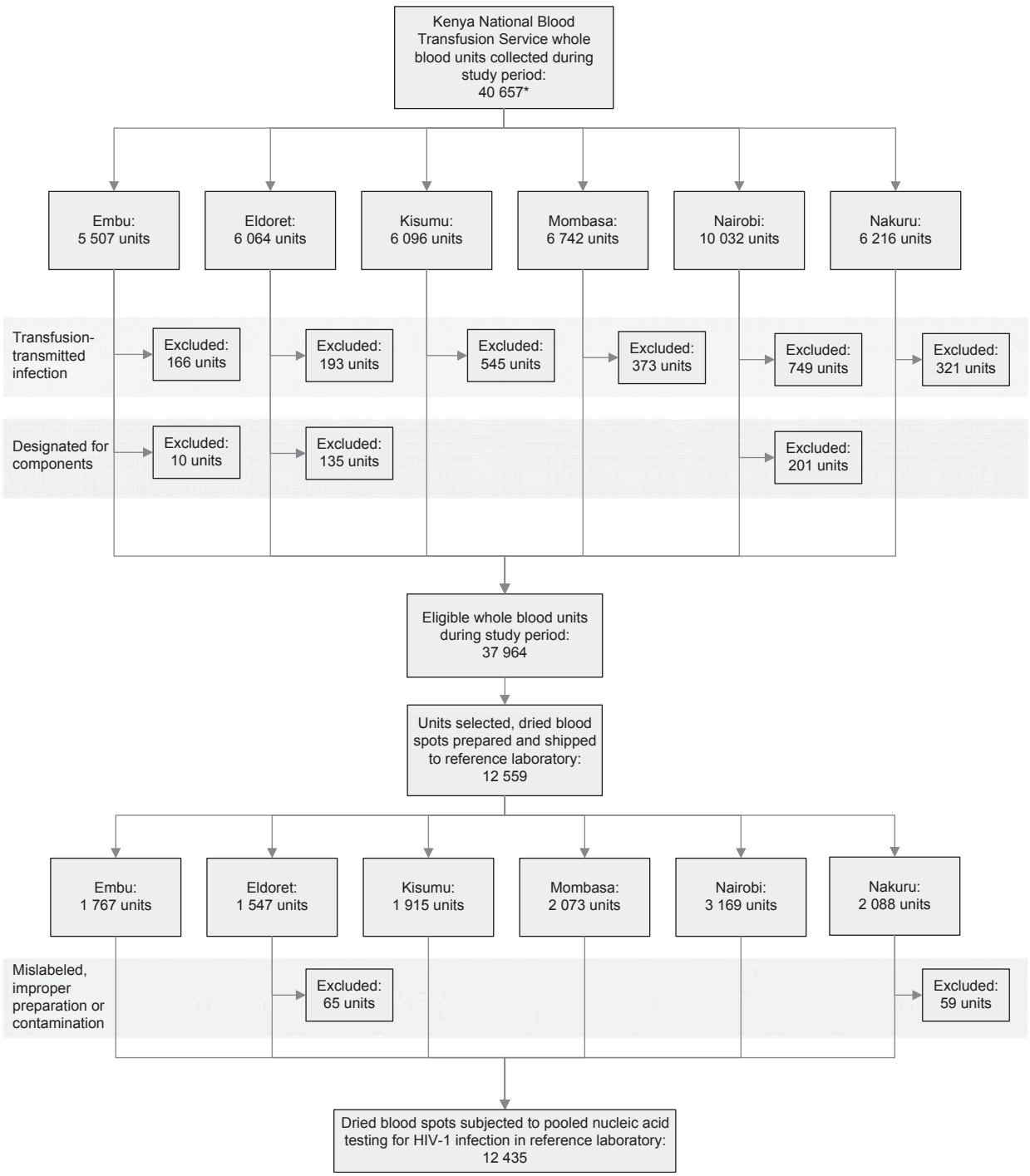


Fig. 1 Study profile of whole blood units collected by Kenya National Blood Transfusion Service and dried blood spots subjected to testing at U.S. Centers for Disease Control and Prevention (CDC) reference laboratory in Kisumu, Kenya. *The study period for the Mombasa centre included November–December, 2008 and for all six centres was March–June, 2009.

In this study, the lack of ELISA window period cases that could have been detected by NAT is of specific importance to resource-limited settings where NAT may not be economically feasible. In sub-Saharan Africa, only South Africa and Namibia perform universal NAT screening for donated blood [26,27]. Elsewhere in the region, one limited evaluation found that NAT could improve safety but would cost three to ten times more than fourth-generation

No claim to original US government works

Table 1 Improvement of blood transfusion safety in Kenya with the implementation of WHO recommendations for national blood programs: 1994 and 2009

	Year study conducted	
	1994[a]	2009
National blood policies and procedures		
Presence of centrally co-ordinated national blood transfusion service	No	Yes
Legislative policy mandating 100% blood collections from voluntary, non-remunerated donors	No	Yes
Universal screening of donor blood for HIV	No	Yes
Laboratory test results		
HIV blood donor prevalence	6·4%	1·2%
Samples with HIV-negative results in blood centres, retested in reference laboratory	1,734	12 435
Samples with HIV-negative results in blood centres with HIV reactive results on subsequent retesting in reference laboratory[b]	41	0
Residual risk of transfusion-associated HIV transmission due to reporting or laboratory error or omission	2·1%	0%

[a]Data obtained from Moore *et al.* [2].

[b]National screening algorithm in 1994 consisted of HIV antibody-only assays and in 2009 was combined fourth-generation p24 antigen and HIV antibody assays. Reference laboratory testing in 1994 consisted of HIV antibody-only testing while in 2009 was pooled nucleic acid testing.

screening [28]. Our findings suggest that NAT would only confer a marginal safety advantage to the Kenya NBTS compared to the current combination strategy of stringent donor selection and screening with fourth-generation assays. In other countries with severe, generalized HIV epidemics, program planners should carefully consider the potential safety advantage and costs before implementing NAT.

One limitation of this study is the lack of inclusion of blood units collected and screened outside of the NBTS network. While the vast majority of blood transfusions in Kenya are estimated to use blood collected and screened by the NBTS, when health care facilities face shortages of blood, especially during school holiday seasons, collection and screening at the hospital level may occur. Additionally, some large private hospitals in the country have voluntarily opted out of the NBTS system. Accurate data on non-NBTS collections and laboratory quality assurance systems are unavailable, but it is generally assumed they are less rigid than NBTS practices. Furthermore, this study does not quantify window period cases undetectable by both ELISA and NAT. The estimates presented here could therefore underestimate the total residual risk of transfusion-associated HIV transmission in the country. Another limitation of this study is the inability to quantify the impact of declining prevalence of HIV in the population on HIV prevalence among blood donors, and the associated residual risk estimate. In 1994, modelling suggests that HIV incidence was at its highest while prevalence, based on trends in antenatal patients, peaked in the late 1990s [29]. Since 2000, prevalence has declined in urban areas but appears to be increasing in rural areas [6]. Although this decline may have an impact on the blood donor population and on the reduction in the residual risk of transfusion-associated HIV infection, we believe these findings support a conclusion that the dramatic improvement in blood safety is more likely a result of investments to strengthen Kenya's blood safety program.

As blood services in sub-Saharan Africa continue to develop, future studies should reassess the safety benefit of implementing NAT. The successes observed in Kenya should encourage other countries in sub-Saharan Africa to fully adopt the WHO Aide-mémoire strategies for blood safety and to provide leadership and commit resources to strengthen their national blood transfusion services. Given the remarkable potential of these strategies to prevent HIV infections, we hope that donor agencies and governmental organizations will support continued technical and financial assistance to achieve or maintain high standards of blood safety and eliminate transfusion-associated HIV infections in the region.

Authors and contributors

SVB, JM, CZ, and LHM oversaw the design and conduct of the study and participated in data analyses. DK, RM, and WS were responsible for conducting the study in the field. CZ and JAO were responsible for testing by the reference laboratory. RWS was responsible for study design and data analysis. JN and SO participated in study design. JPP participated in study design and data interpretation. All authors took part in the preparation of the article and approved the final version.

Funding

U.S. President's Emergency Plan for AIDS Relief (PEPFAR). Funding was provided by PEPFAR to CDC and NBTS. CDC and NBTS staff participated in the design, data collection, analysis, and interpretation of the data, as well as in writing the manuscript and the decision to submit it for publication.

Conflicts of interest

We declare that we have no conflicts of interest.

No claim to original US government works

Disclaimer

The findings and conclusions in this report are those of the authors and do not necessarily represent the views of the U.S. Centers for Disease Control and Prevention. Use of trade names is for identification purposes only and does not constitute endorsement by the U.S. Centers for Disease Control and Prevention or the Department of Health and Human Services.

Acknowledgements

We thank the site co-ordinators and laboratory staff of each of the NBTS regional centres and Charles Rombo; the laboratory staff of the KEMRI/CDC Kisian Field Station reference laboratory; and Reinhard Kaiser, Janet Moore, Chin-Yih Ou, Bharat Parekh, and Richard Respess for critical review and advice.

This paper is published with the approval of the Director KEMRI. [Acknowledgement added here after initial online publication.]

References

1 World Health Organization. Aide-Memoire for National Blood Programmes. Geneva, Switzerland, World Health Organization, 2002. Available at http://www.who.int/bloodsafety/transfusion_services/en/Blood_Safety_Eng.pdf

2 Moore A, Herrera G, Nyamongo J, Lackritz E, Granade T, Nahlen B, Oloo A, Opondo G, Muga R, Janssen R: Estimated risk of HIV transmission by blood transfusion in Kenya. *Lancet* 2001; **358**:657–660

3 Kenya Ministry of Health. National Blood Safety Policy. Ministry of Health, 2001

4 U.S. President's Emergency Plan for AIDS Relief. The power of partnerships: 2008 Annual Report to Congress. Washington, DC, 2008. Available at http://www.pepfar.gov/documents/organization/100029.pdf

5 CDC. Progress toward strengthening blood transfusion services – 14 countries, 2003–2007. *Morb Mortal Wkly Rep* 2008; **57**: 1273–1277

6 National AIDS/STI Control Programme (NASCOP), Kenya. *2007 Kenya AIDS Indicator Survey: Final Report.* Nairobi, NASCOP. September 2009

7 Ly TD, Ebel A, Faucher V, Fihman V, Laperche S: Could the new HIV combined p24 antigen and antibody assays replace p24 antigen specific assays? *J Virol Methods* 2007; **143**:86–94

8 Le Corfec E, Le Pont F, Tuckwell HC, Rouzioux C, Costagliola D: Direct HIV testing in blood donations: variation of the yield with detection threshold and pool size. *Transfusion* 1999; **39**:1141–1144

9 Ikomey GM, Atashili J, Okomo-Assoumou MC, Mesembe M, Ndumbe PM: Dried blood spots versus plasma for the quantification of HIV-1 RNA using the manual (PCR-ELISA) amplicor monitor HIV-1 version 1.5 assay in Yaounde, Cameroon. *J Int Assoc Physicians AIDS care* 2009; **8**:181–184

10 Alvarez-Muñoz MT, Zaragoza-Rodríguez S, Rojas-Montes O, Palacios-Saucedo G, Vázquez-Rosales G, Gómez-Delgado A, Torres J, Muñoz O: High correlation of human immunodeficiency virus type-1 viral load measured in dried-blood spot samples and in plasma under different storage conditions. *Arch Med Res* 2005; **36**:382–386

11 Reigadas S, Schrive MH, Aurillac-Lavignolle V, Fleury HJ: Quantitation of HIV-1 RNA in dried blood and plasma spots. *J Virol Methods* 2009; **161**:177–180

12 Kane CT, Ndiaye HD, Diallo S, Ndiaye I, Wade AS, Diaw PA, Gaye-Diallo A, Mboup S: Quantitation of HIV-1 RNA in dried blood spots by the real-time NucliSENS EasyQ HIV-1 assay in Senegal. *J Virol Methods* 2008; **148**:291–295

13 Monleau M, Montavon C, Laurent C, Segondy M, Montes B, Delaporte E, Boillot F, Peeters M: Evaluation of different RNA extraction methods and storage conditions of dried plasma or blood spots for human immunodeficiency virus type 1 RNA quantification and PCR amplification for drug resistance testing. *J Clin Microbiol* 2009; **47**:1107–1118

14 Schreiber GB, Busch MP, Kleinman SH, Korelitz JJ: The retrovirus epidemiology donors study. The risk of transfusion-transmitted viral infections. *N Engl J Med* 1996; **334**:1685–90

15 Lackritz EM, Satten GA, Aberle-Grasse J, Dodd RY, Raimondi VP, Janssen RS, Lewis WF, Notari EP, Petersen LR: Estimated risk of transmission of the human immunodeficiency virus by screened blood in the United States. *N Engl J Med* 1995; **333**:1721–1725

16 Stover J: AIM: A Computer Program for Making HIV/AIDS Projections and Examining the Social and Economic Impact of AIDS. Glastonbury, CT: Futures Institute, December 2007. Available at http://www.FuturesInstitute.org

17 Ouattara H, Siransy-Bogui L, Fretz C, Diane KM, Konate S, Koidio A, Minga KA, Hyda J, Koffi-Abe N, Offoumou AM, Abissey S: Residual risk of HIV, HBV, and HCV transmission by blood transfusion between 2002 and 2004 at the Abidjan National Blood Transfusion Center. *Transfus Clin Bio* 2006; **2250**:1–4

18 Clopper C, Pearson ES: The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika* 1934; **26**:404–413

19 Consten EC, van der Meer JT, de Wolf F, Heij HA, Henny PC, van Lanschot JJ: Risk of iatrogenic human immunodeficiency virus infection through transfusion of blood tested by inappropriately stored or expired rapid antibody assays in a Zambian hospital. *Transfusion* 1997; **37**:930–934

20 McFarland W, Mvere D, Katzenstein D: Risk factors for prevalent and incident HIV infections in a cohort of volunteer blood donors in Harare, Zimbabwe: implications for blood safety. *AIDS* 1997; **11**(Suppl 1):S97–102

21 Baggaley RF, Boily MC, White RG, Alary M: Risk of HIV-1 transmission for parenteral exposure and blood transfusion: a systematic review and meta-analysis. *AIDS* 2006; **20**:805–812

22 Gisselquist D, Perrin L, Minkin SF: Parallel and overlapping HIV and bloodborne hepatitis epidemics in Africa. *Int J STD AIDS* 2004; **15**:145–152

No claim to original US government works

23 McFarland W, Mvere D, Shandera W, Reingold A: Epidemiology and prevention of transfusion-associated human immunodeficiency virus transmission in sub-Saharan Africa. *Vox Sang* 1997; **72**:85–92

24 Sarkodie F, Adarkwa M, Adu-Sarkodie Y, Candotti D, Acheampong JW, Allain JP: Screening for viral markers in volunteer and replacement blood donors in West Africa. *Vox Sang* 2001; **80**:142–147

25 van der Poel CL, Seifried E, Schaasberg WP: Paying for blood donations: still a risk? *Vox Sang* 2002; **83**:285–293

26 Heyns Adu P, Benjamin RJ, Swanevelder JP, Laycock ME, Pappalardo BL, Crookes RL, Wright DJ, Busch MP: Prevalence of HIV-1 in blood donations following implementation of a structured blood safety policy in South Africa. *JAMA* 2006; **295**:519–526

27 Fang CT, Field SP, Busch MP, Heyns AP: Human immunodeficiency virus-1 and hepatitis C virus RNA among South African blood donors: estimation of residual transfusion risk and yield of nucleic acid testing. *Vox Sang* 2003; **85**:9–19

28 van Hulst M, Sagoe KW, Vermande JE, van der Schaaf IP, van der Tuuk Adriani WP, Torpey K, Ansah J, Mingle JA, Smit Sibinga CT, Postma MJ: Cost-effectiveness of HIV screening in blood donations in Accra (Ghana). *Value Health* 2008; **11**:809–819

29 Cheluget B, Baltazar G, Orege P, Ibrahim M, Marum LH, Stover J: Evidence for population level declines in adult HIV prevalence in Kenya. *Sex Transm Infect* 2006; **1**:i21–i26

No claim to original US government works